BIO 004 · HUMAN ANATOMY

Lab Structure List.

Every structure you are responsible for on the models, the specimens and the slides, all five units in one place. This is the list your lab practicals are built from. Each unit opens with the sprints that drill it, so you can go from a name on this page to practicing it. Check a structure off when you can find it without looking it up, then again a week later without your notes.

1 **Unit 1. Foundations, Tissues and the Integument**

2 **Unit 2. The Skeletal System and Joints**

3 **Unit 3. Muscle, the Heart and the Vessels**

4 **Unit 4. Limb Muscles, Respiratory and Digestive**

5 **Unit 5. Urinary, Reproductive and the Nervous System**

UNIT 1

Foundations, Tissues and the Integument

TERMINOLOGY, THE MICROSCOPE, TISSUES AND THE INTEGUMENT

The safety sheet, the microscope, and the first structures you identify. Work it with the models and the slides in front of you.

☐ **Drill these on the models**

LAB SPRINTS FOR THIS UNIT

Anatomical Terminology

Body Cavities and Regions

Tissues and Histology

Integumentary System

☐ ANATOMY LABORATORY SAFETY SHEET

☐ Student supplies/materials

Each student will need to provide the following:

Lab coat. Lab coats must be worn to protect street clothes from contacting laboratory materials

☐ Nitrile gloves

☐ Protective eyewear

Note that students will store these items in the anatomy lab during the semester

☐ General Lab Rules

No cameras. Electronic devices (phones, tablets, smart watches, laptops) with cameras may not be used in the anatomy laboratory. Place electronic devices with cameras in your bag while you are in the lab

No phones or other wireless electronic devices. Set your phone or other wireless device (tablets, smart watches, laptops, etc.) to silent and place it in your bag while you are in the lab

No personal sound systems, Phones, MP3 players, etc. may not be used in the anatomy laboratory. Place these devices in your bag while you are in the lab

No food and drink. Never put anything into your mouth while in the laboratory. Do not drink, eat, or chew gum. Do not bring any food or drink into the laboratory. Do not apply makeup esp. around your mouth (including lip balm & lip gloss)

Proper shoes and clothing - solid-toed shoes are required. Your feet must be protected from physical and chemical harm. Wear long pants on days when preserved specimens are used

Keep the lab organized. Return all materials (models, specimens, and microscope slides) to their proper location. This is typically where you received them or as designated by your instructor. Microscope slides are to be cleaned (if necessary) and returned to their proper tray

☐ Keep the lab clean

Under no circumstances should materials from the wet lab be moved to the dry lab

Under no circumstances should materials from the dry lab be moved to the wet lab

Do not touch clean materials if your hands are contaminated with wetting solution

Whenever you are wearing gloves, assume your hands are contaminated

Worktops are to be wiped clean at the end of each class

Laboratory Safety Procedures Specific to the Handling of Human Cadavers

Wear a lab coat, gloves, and long pants at all times in the wet lab

Preserved specimens are used in ventilated areas exclusively (on cadaver tables or ventilated worktops)

Dispose of all contaminated papers and gloves in the appropriate receptacle

Place used scalpel blades in sharps containers as soon as a procedure is completed

In the event of a spill or an injury, notify the instructor immediately

Any student observing a violation of the above rules is to notify the instructor immediately so that the potential resulting danger can be avoided

If you are pregnant or may become pregnant you must speak with and provide documentation that you have spoken with, your physician/obstetrician concerning the potential dangers of working in a lab with formaldehyde and phenol while pregnant

------------------------------Detach and return lower portion to instructor ------------------------------

I have read and understand the rules outlined in the Anatomy Laboratory Safety Sheet (above) and agree to abide by them

☐ Printed Name Signature Date ☐ ______________ ☐ - blank page -

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WORKSHEET 1.0

Anatomical Position, Directional Terms, and Planes of Section

☐ Anatomical Position

Working with a partner, stand in anatomical position. How is anatomical position different from the typical standing position?

Brad is modeling the anatomical position to his lab partner Tim. He is looking down at his feet, his palms are facing backward, and his feet are flat on the floor and parallel. Is he correctly modeling the anatomical position? If not, describe what he is doing wrong

Why is the anatomical position important in the study of the human body?

☐ Directional Terminology

Define the following terms use them in a sentence

Superior / Cranial* / Rostral*	______________
Inferior / Caudal*	______________
Anterior / Ventral*	______________
Posterior / Dorsal*	______________
Medial	______________
Lateral	______________
Proximal	______________
Distal	______________
Deep	______________
Superficial	______________

Fill in the blank with the correct directional terminology

☐ The skin is ______________ to the skeleton

☐ The vertebral column is ______________ to the heart

☐ The nose is ______________ to the ears (auricles)

The abdominal cavity is ____________________ to the pelvic cavity

☐ Body Planes and Sections

Draw the body as it appears when sectioned in each of the three different planes

Transverse Section Frontal Section Sagittal Section

WORKSHEET 1.1

The Compound Microscope, Keep this worksheet for study

☐ **Care of the Compound Microscope**

What is the proper method for transporting and storing the microscope?

What can you use to clean the objective lenses or ocular lenses? ____________________

☐ **Structure of the Compound Microscope**

Be able to define and locate the following terms of the Compound Microscope

Revolving nosepiece Lever to adjust Iris Diaphragm

☐ Scanning Objective Lens (4×) Condenser
☐ Low Power Objective Lens (10×) Field of View
☐ High Power Objective Lens (40×) Stage

High "Dry" Power Objective lens (100×) Eyepiece w/ Ocular Lens

☐ Coarse adjustment knob Arm
☐ Fine adjustment knob Lamp
☐ Short Answer Questions

What is the microscope field of view (FOV)?

What is the relationship between the size of the field and the objective used to observe the field?

Which of the following are TRUE or FALSE?

Circle the correct answer. If the statement is false, correct it by writing the proper word or phrase to replace the word or phrase that is underlined

(TRUE / FALSE) When the scanning objective is used the total magnification is 10×

(TRUE / FALSE) The microscope is stored with the high power lens in position

(TRUE / FALSE) Always begin focusing with the scanning objective lens

(TRUE / FALSE) The coarse adjustment knob may be used in focusing with all four objectives

(TRUE / FALSE) The iris diaphragm is a lens used to concentrates the light on the specimen

(TRUE / FALSE) The ocular is the lens in the eyepiece, which has a magnification of 10× (in this lab)

(TRUE / FALSE) The condenser controls the width of the circle of light passing through the specimen

(TRUE / FALSE) The black ring on the revolving nosepiece is always used when changing objectives

(TRUE / FALSE) The microscope is stored with the stage moved to its lowest position

☐ Viewing objects through the microscope
☐ Obtain a letter e slide

Sketch the letter e in the circle just as it appears in the field

Scanning Power Objective Lens	**Low Power Objective Lens**	**High Power Objective Lens**
____________________	____________________	____________________
Magnification of objective lens: ______________x	**Magnification of objective lens:** ______________x	**Magnification of objective lens:** ______________x
Total magnification of microscope: ______________x	**Total magnification of microscope:** ______________x	**Total magnification of microscope:** ______________x

How has the orientation of the e changed from the slide to the view in the microscope field?

When viewing the letter e slide, if moving the slide to the left on the stage as you view it through the ocular lens, in what direction does the image move? ____________________

When viewing the letter e slide, if moving the slide to the right on the stage as you view it through the ocular lens, in what direction does the image move? ____________________

When viewing the letter e slide, if moving the slide away from you on the stage as you view it through the ocular lens, in what direction does the image move? ____________________

Why is it necessary to center the object you wish to view before changing to a higher power objective lens?

☐ **Perceiving Depth**

Obtain a cross-thread slide and observe it under the microscope

☐ Which thread is lowest? _____________

☐ Which thread is in the middle? _____________

☐ Which thread is highest? _____________

Determining the diameter of the microscope Field of View (F.O.V.)

Use a clear plastic ruler to measure the field diameter with the scanning power objective. The ruler does not have 0.1mm marks, do your best to estimate to the diameter to the nearest tenth of a mm

Record your measurement on the table below and complete the rest of the Scanning Power column

Now calculate the field diameter for the low, high and high dry lenses using the following formula:

Record the calculated field diameter in the table below. Complete the table below

	SCANNING POWER	LOW POWER	HIGH POWER	HIGH "DRY"
Magnification of Objective				
Total Magnification				
F.O.V. (also called the field diameter)	Measure with ruler mm	Calculate w/ formula mm	Calculate w/ formula mm	Calculate w/ formula mm
F.O.V. in microns	μm	μm	μm	μm

☐ **Estimate the diameter of a red blood cell**

Obtain a blood smear slide. Observe the slide under the microscope. Use the appropriate objective to view the slide

Using the information you calculated in the activity above, estimate the diameter of a red blood cell

Hint: Estimate the number of red blood cells that could fit across the F.O.V. if placed end-to-end when using the "high dry" 100× objective. Divide the field diameter by the number of red blood cells that can fit across the F.O.V. This will give you the approximate size of a single red blood cell

What is your estimate of the size of a red blood cell? Express your answer using the correct units

Your estimate of the size of an RBC ____________________. Write this number (and your initials) to the board

When you are finished using a microscope always:

Turn the revolving nosepiece to the scanning objective

☐ Lower the stage to its lowest position

☐ Dim the lamp then turn it off

In most classes returning the microscope to its cabinet is standard. For this class please cover replace the dust cover and gently slide the microscope away from the table edge

WORKSHEET 1.2

Epithelial Tissue

☐ Introduction to Epithelial Tissue

Note: You will need to refer to your textbook and lecture notes to complete the following questions

Q 1.1 List the six characteristics of epithelial tissues that distinguish them from other tissue types

Q 1.2 Describe how epithelial tissue is classified

☐ 2. Simple Epithelia

Using the microscope, examine each of the simple epithelial tissues in the table below

In the boxes provided, carefully draw each tissue as you see it under the microscope

Using your textbook or histology book, label the apical surface, basement membrane, lamina propria (if present), nuclei, and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give an example of where in the body you would find each type of epithelium

SIMPLE SQUAMOUS EPITHELIUM	SIMPLE CUBOIDAL EPITHELIUM	SIMPLE COLUMNAR EPITHELIUM	PSEUDOSTRATIFIED COLUMNAR EPITHELIUM
Slide:	Slide:	Slide:	Slide:
	microvilli (if present)	microvilli, goblet cells	cilia, goblet cells

Q 2.1 Where is simple squamous epithelium located in the body? What is the function of this tissue?

Q2.2 Where is simple cuboidal epithelium located in the body? What is the general function of this tissue?

Q2.3 Where is simple columnar epithelium located in the body? What is the function of this tissue?

Q2.4 Why is pseudostratified epithelium a type of simple epithelium not a type of stratified epithelium?

☐ 3. Stratified Epithelia

Using the microscope, examine each of the stratified epithelial tissues in the table below

In the boxes provided, carefully draw each tissue as you see it under the microscope

Using your textbook or histology book, label the apical surface, basement membrane, lamina propria (if present), nuclei, and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give an example of where in the body you would find each type of epithelium

STRATIFIED SQUAMOUS EPITHELIUM, NON-KERATINIZED	STRATIFIED SQUAMOUS EPITHELIUM, KERATINIZED	TRANSITIONAL EPITHELIUM
Slide:	Slide:	Slide:
Estimate number of cell layers	Estimate number of cell layers	Estimate number of cell layers

Q 3.1 Where is transitional epithelium located in the body? What is the function of this tissue?

Q 3.2 Where is stratified squamous epithelium located in the body? What is the function of this tissue?

Q 3.3 How does transitional epithelium differ from other stratified epithelia?

WORKSHEET 1.3

Connective Tissue

☐ Introduction to Connective Tissue

Note: Refer to your textbook (esp. Chapter 4) and lecture notes to complete the following questions

Q 1.1 List the four main classes of connective tissue?

Q 1.2 Describe the general organization or basic structure of connective tissues

☐ Loose Connective Tissue

Using the microscope, examine each of the loose connective tissues in the table below

In the boxes provided, carefully draw each tissue as you see it under the microscope

Using your textbook or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give an example of where in the body you would find each type of connective tissue

LOOSE AREOLAR C.T	ADIPOSE C.T	RETICULAR C.T
Slide:	Slide:	Slide:
collagen fibers, elastic fibers, fibroblasts, mast cells	adipocytes (fat cells), fat vacuoles, blood vessels	reticular fibers

Q 2.1 Where is loose areolar connective tissue located in the body? What is the function of this tissue?

Q2.2 Where is adipose tissue located in the body? What is the function of this tissue?

Q2.3 Where is reticular connective tissue located in the body? What is the function of this tissue?

☐ 3. Dense Connective Tissue

Using the microscope, examine each of the dense connective tissues in the table below

In the boxes provided, carefully draw each tissue as you see it under the microscope

Using your textbook or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give an example of where in the body you would find each type of connective tissue

☐ Dense Regular C.T	☐ Slide:	☐ collagen fibers, fibroblasts
☐ Dense Irregular C.T	☐ Slide:	☐ collagen fibers, fibroblasts
☐ Elastic C.T	☐ Slide:	☐ elastin fibers

Q 3.1 Compare and contrast the structure and function of dense irregular connective tissue with the structure and function of dense regular connective tissue

Q 3.2 Where is elastic connective tissue located in the body? What is the function of this tissue?4. Cartilage and Osseous Tissue

Using the microscope, examine each of the dense connective tissues in the table below

In the boxes provided, carefully draw each tissue as you see it under the microscope

Using your textbook or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give an example of where in the body you would find each type of cartilage

HYALINE CARTILAGE	ELASTIC CARTILAGE	FIBROCARTILAGE	OSSEOUS TISSUE
Slide:	Slide:	Slide:	Slide:
perichondrium, lacunae, chondrocytes, extracellular matrix	perichondrium, lacunae, chondrocytes, extracellular matrix, elastin fibers	lacunae, chondrocytes, extracellular matrix, collagen fibers Note: no pericondrium!	osteons, central canal, lacunae, lamellae, osteocytes, canaliculi

Q 4.1 List two locations for each of the cartilages listed above

Q 4.2 What is the function of each of the cartilages listed above?

Q 4.3 How does the structural organization of connective tissue differ from epithelial tissue?

Q 4.4 Explain how bone tissue is different from a bone (such as the femur)

WORKSHEET 1.4

Muscle Tissue

☐ Skeletal Muscle Tissue

Using the microscope, examine skeletal muscle on the slides indicated by your instructor

In the boxes provided, carefully draw this tissue as you see it under the microscope

Using your textbook or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give examples of where in the body you would find skeletal muscle tissue

SKELETAL MUSCLE (DRAWN FROM IMAGES / FIGURES IN CHAPTER 10 OF YOUR TEXTBOOK)	SKELETAL MUSCLE (LOOKING THROUGH MICROSCOPE)
Skeletal Muscle (drawn from images / figures in chapter 10 of your textbook)	Slide:
In both drawings above, label: skeletal muscle fibers, sarcomere, sarcolemma, Z disc, A band, I band, M line, actin and myosin. Note: some features will be too small to see when looking through the microscope, in this case label the locations where you would find these structures	In both drawings above, label: skeletal muscle fibers, sarcomere, sarcolemma, Z disc, A band, I band, M line, actin and myosin. Note: some features will be too small to see when looking through the microscope, in this case label the locations where you would find these structures

Q 1.1 Define the following terms:

Muscle fiber:

Endomysium:

Fascicle:

Perimysium:

Sarcomere:

Multinucleate:

Q 1.2 What is a striation? Note the striations in the skeletal muscle fibers

☐ 2. Cardiac Muscle Tissue

Using the microscope, examine cardiac muscle on the slide indicated by your instructor

In the box provided, carefully draw this tissue as you see it under the microscope

Using your textbook or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give an example of where in the body you would find cardiac muscle tissue

Q 2.1 Define intercalated disc. Note the difference in appearance between intercalated discs and striations in cardiac muscle fibers

Q 2.2 How is cardiac muscle tissue different from skeletal muscle tissue?

☐ 3. Smooth Muscle Tissue

Using the microscope, examine smooth muscle on the slide(s) indicated by your instructor

In the boxes provided, carefully draw this tissue as you see it under the microscope

Using your textbook or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give examples of where in the body you would find smooth muscle tissue

Q 3.1 How is the appearance of smooth muscle tissue different if you observe a cross-sectional view of the tissue rather than a longitudinal view?

Q3.2 How is smooth muscle tissue different from skeletal muscle tissue and how is it different from cardiac muscle tissue?

CARDIAC MUSCLE	SMOOTH MUSCLE
Slide:	Slide:
cardiac muscle fibers, A band, I band, intercalated discs	smooth muscle fibers

WORKSHEET 1.5

The Microscopic Structure of Nervous Tissue

- [ ] **Nervous Tissue**

Using the microscope, examine each of the nervous tissue slides in the table below

In the boxes below, draw each example of nervous tissue as you see it under the microscope

Using your textbook (chapters 4 and 12) or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

Be able to give examples of where in the body you would find examples of nervous tissue

MULTIPOLAR NEURONS	SPINAL CORD	NERVE (L.S.& C.S.)
Slide:	Slide:	Slide:
multipolar neurons, cell bodies, nucleolus, cell processes (including dendrites, axons, and axon hillocks), glial cells	white matter, gray matter, ventral horn, dorsal horn, multipolar neurons	axons, Schwann cells, nodes of Ranvier, epineurium, perineurium, endoneurium Note: C.T. structures are best seen in cross section

Q1.1 What is the function of nervous tissue?

Q1.2 Why are neuroglia found in nervous tissue?

Q1.3 In what ways are white matter and gray matter different?

Q1.4 What is myelin? Why are some neuronal processes myelinated?

Q1.5 What is a node of Ranvier? What is its function?

WORKSHEET 1.6

The Microscopic Structure of Blood Vessels and Formed Elements

- [ ] **Blood Vessels**

Using the microscope, examine the slide showing a cross section of an artery, vein, and nerve

In the boxes below, draw each organ as you see it under the microscope

Using your textbook (first several pages of chapter 20) or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed at the bottom of each box or described by instructor)

ARTERY (CROSS SECTION)	VEIN (CROSS SECTION)	NERVE (CROSS SECTION)
Slide:	Slide:	Slide:
tunica intima (with endothelium and internal elastic lamina) tunica media (with smooth muscle and dense elastic connective tissue) tunica externa lumen, erythrocytes	tunica intima (with endothelium) tunica media (with smooth muscle) tunica externa lumen, erythrocytes	Compare to blood vessels and note proximity to blood vessels. Identify as many of the structures as possible that you observed in worksheet 1.5

- [ ] Q1.1 Define artery
- [ ] Q1.2 Define vein

Q1.3 What is the function of the smooth muscle in the tunica media?

Q1.4 What is the function of the dense elastic tissue in arteries?

Q1.5 Compare the structure of an artery and vein. What about them is different? What is similar?

- [ ] **The Formed Elements of Blood**

Using the microscope, examine the slide showing a blood smear

In the boxes below, draw each cell as you see it under the microscope

Use your textbook (first several pages of chapter 18) to identify each of the formed elements and describe their primary functions

COMPOSITION OF BLOOD PLASMA FORMED ELEMENTS ERYTHROCYTES THROMBOCYTES LEUKOCYTES NEUTROPHILS (60% OF WBCS) LYMPHOCYTES (30% OF WBCS) MONOCYTES (6% OF WBCS) EOSINOPHILS (3% OF WBCS) BASOPHILS (1% OF WBCS)

The granulocytes: leukocytes that all have distinct granules, and a bi- or multi-lobed nucleus

The agranulocytes: leukocytes that do not have distinct granules

WORKSHEET 1.7

The Integumentary System

☐ Thick Skin

Using the microscope, examine the slide showing a section of thick skin

In the box below, draw the structures you see it under the microscope

Using your textbook (chapter 5) or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed to the left of each layer or described by instructor)

Also be able to identify the structures below on the skin model in the lab

THICK SKIN	
	Slide:
Layer: Epidermis Tissue: stratified squamous epithelial tissue Cells: keratinocytes Sublayers: stratum corneum stratum lucidum* stratum granulosum stratum spinosum stratum basale melanocytes basement membrane * thick skin only	
Layer: Dermis Tissues: areolar C.T. and dense irregular C.T. Cells: fibroblasts Sublayers: papillary (sub)layer areolar C.T. dermal papillae Meissner's (tactile) corpuscles reticular (sub)layer dense irregular C.T. Pacinian (lamellar) corpuscles	
Layer: Hypodermis* Tissues: adipose C.T. and areolar C.T. Cells: fibroblasts, adipocytes * Not part of integumentary system; also called superficial fascia or subcutaneous layer	

Q1.1 Describe the main functions of the epidermis, dermis and hypodermis

☐ Thin Skin

Using the microscope, examine the slide showing a section of then skin

In the box below, draw the structures you see it under the microscope

Using your textbook (chapter 5) or histology book, label nuclei and any other special structures or cells associated with specific tissues (listed to the left of each layer or described by instructor)

Also be able to identify the structures below on the skin model in the lab

THIN SKIN	
	Slide:
Layer: Epidermis Tissue: stratified squamous epithelial tissue Cells: keratinocytes Sublayers: stratum corneum stratum granulosum stratum spinosum stratum basale	
Layer: Dermis Tissues: areolar C.T. and dense irregular C.T. Cells: fibroblasts Sublayers: papillary (sub)layer areolar C.T. dermal papillae reticular (sub)layer dense irregular C.T. Appendages of the skin: Sweat gland (eccrine gland) pore, secretory cells, duct hair root, shaft hair follicle hair bulb, arrector pili muscle sebaceous gland duct, secretory cells (secrete sebum)	
Layer: Hypodermis* Tissues: adipose C.T. and areolar C.T. Cells: fibroblasts, adipocytes * normally not considered part of the skin	

Q 2.1 What type of muscle tissue is the arrector pili muscle?

Q2.2 How does a holocrine gland secrete its product?

UNIT 2

The Skeletal System and Joints

BONES, BONE MARKINGS AND THE JOINTS

Every bone and bone marking you are responsible for, plus the joints. This is the largest naming load in the course.

- [ ] **Drill these on the models**

LAB SPRINTS FOR THIS UNIT

Bone Microanatomy
Axial Skeleton: The Skull
Axial Skeleton: Spine and Thoracic Cage
Appendicular Skeleton: Upper Limb
Appendicular Skeleton: Lower Limb

WORKSHEET 2.1

The Appendicular Skeleton I, Bones of the Pectoral Girdle and Upper Limbs

Identifying the bones and bone markings on the appendicular skeleton

- [ ] **A. Pectoral girdle**
- [ ] **1. Clavicle**

- [ ] sternal end
- [ ] acromial end
- [ ] conoid tubercle

- [ ] **2. Scapula (plural: scapulae)**

- [ ] acromion
- [ ] glenoid cavity
- [ ] superior border
- [ ] coracoid process
- [ ] supraglenoid tubercle
- [ ] superior angle
- [ ] scapular spine
- [ ] infraglenoid tubercle
- [ ] suprascapular notch
- [ ] infraspinous fossa
- [ ] axillary (lateral) border
- [ ] supraspinous fossa
- [ ] vertebral (medial) border
- [ ] subscapular fossa
- [ ] inferior angle

- [ ] **B. Bones of the upper limbs**
- [ ] **1. Humerus (plural: humeri)**

HEAD	DELTOID TUBEROSITY	TROCHLEA
anatomical neck	radial groove*	coronoid fossa
surgical neck	lateral epicondyle	capitulum
greater tubercle	lateral supracondylar ridge	radial fossa
lesser tubercle	medial epicondyle	olecranon fossa
intertubercular sulcus (bicipital groove)	medial supracondylar ridge	

- [ ] **2. Radius (plural: radii) head and neck styloid process interosseous crest radial tuberosity ulnar notch**
- [ ] **3. Ulna (plural: ulnae)**

- [ ] head styloid process interosseous crest
- [ ] ulnar tuberosity* radial notch

coronoid process trochlear notch olecranon process

- [ ] **4. Wrist, hand and digits**

a. Carpals (identify each on an articulated wrist)

Memory device for the carpals, So long to pinky, here comes the thumb

Proximal row from lateral to medial, So long to pinky

- [ ] **scaphoid, lunate, triangular, pisiform**

Distal row from medial to lateral, Here comes the thumb

- [ ] **hamate, capitate, trapezoid, trapezium**

b. Metacarpals (numbered 1–5 lateral to medial. Identify each on an articulated hand) c. Phalanges (singular is phalanx). Distal, middle (except 1st digit) and proximal phalanges. (Note: 1st digit is also called the pollex)

WORKSHEET 2.2

The Appendicular Skeleton II, Bones of the Pelvic Girdle and Lower Limbs

- [ ] **A. The Pelvic Girdle**
- [ ] **1. Coxal bone (also called os coxa or pelvic bone or hip bone or the innominate bone)**
- [ ] **a. Ilium (pural. ilia)**

ILIAC CREST	ANTERIOR SUPERIOR ILIAC SPINE (A.S.I.S)
auricular surface	anterior inferior iliac spine (A.I.I.S.)
iliac fossa	posterior superior iliac spine (P.S.I.S)
greater sciatic notch	posterior inferior iliac spine (P.I.I.S)

- [ ] ala
- [ ] b. Ischium (plural: ischia)
- [ ] ischial spine lesser sciatic notch
- [ ] ischial tuberosity ramus of the ischium
- [ ] c. Pubis (plural: pubes or pubic bones)

PUBIC BODY	PUBIC SYMPHYSIS – JOINT THAT JOINS THE PUBIC BONES
superior ramus	pubic crest
inferior ramus	pubic tubercle

- [ ] d. Acetabulum
- [ ] e. Obturator foramen

f. Male and female pelves have distinct differences in morphology (form). Compare the male pelvis to the female pelvis for differences in: bone thickness, pubic angle, pelvic inlet, pelvic outlet, acetabulum, ischial spine, ischial tuberosity, iliac spines

- [ ] **B. Lower Limbs**
- [ ] **1. Femur (plural: femurs or femora)**

HEAD	INTERTROCHANTERIC CREST
fovea capitis	gluteal tuberosity*
neck	linea aspera
greater trochanter	lateral and medial supra condylar lines
lesser trochanter	lateral and medial epicondyles
intertrochanteric line*	lateral and medial condyles
patellar surface	intercondylar fossa
adductor tubercle*	

- [ ] **2. Patella (plural: patellas or patellae)**
- [ ] **3. Tibia (plural: tibias or tibiae)**
- [ ] lateral condyle
- [ ] fibular notch
- [ ] medial condyle
- [ ] anterior crest (border)
- [ ] intercondylar eminence
- [ ] medial malleolus
- [ ] tibial tuberosity
- [ ] **4. Fibula (plural: fibulas or fibulae)**
- [ ] **head of fibula lateral malleolus**
- [ ] **5. Tarsals (ankle bones – they also contribute to the heel and proximal half of the foot)**

a. Talus, calcaneus, navicular, medial cuneiform, intermediate cuneiform, lateral cuneiform, cuboid. Memory device for the tarsals -The three C's surround the N and T

- [ ] **6. Metatarsals. Metatarsals are numbered I to V (from medial to lateral)**
- [ ] **7. Phalanges. As per the bones of the digits of the hand. The first digit (called the hallux or great toe) has only a proximal and a distal phalanx. Toes 2, 5 have a proximal, middle, and distal phalanx**

Complete the following questions about the appendicular skeleton

What forearm bone contributes most to the elbow joint? ____________________

What bone of the pectoral girdle articulates anteriorly with the sternum? ____________________

- [ ] **What bone articulates with the acetabulum?** ______________

If a person has a “broken finger”, which bone(s) could be fractured? ____________________

How do the three arches of the foot help distribute the body’s weight?

Which bones are found in the leg? (hint: The leg is not the same as the lower limb)?

To which tarsal does the calcaneal (“Achilles”) tendon attach? ____________________

WORKSHEET 2.3

The Axial Skeleton I, Vertebral Column and Thoracic Cage

Identifying the special features of the vertebral column

☐ **Regions and curvatures**

REGIONS	NORMAL CURVATURES	ABNORMAL CURVATURES
cervical (C1– C7)	cervical curvature (concave posteriorly)	Scoliosis
thoracic (T1 - T12)	thoracic curvature (concave anteriorly)	Kyphosis
lumbar (L1 - L5)	lumbar curvature (concave posteriorly)	Lordosis
sacral (S1, S5)	sacral curvature (concave anteriorly)	
coccygeal		

☐ Intervertebral discs
☐ nucleus pulposus anulus fibrosis
☐ Identifying the bones of the vertebral column

☐ **A. Structure of the typical vertebra**

☐ body (centrum) lamina
☐ pedicles spinous process
☐ vertebral foramen transverse processes

superior articular process & facet vertebral arch inferior articular process & facet intervertebral notch / foramen

☐ **B. Regional characteristics**

☐ **a. Cervical vertebrae, C1– C7**

All cervical vertebrae possess transverse foramina. Typically with a small, oval shaped centrum. The vertebral foramen is typically triangular in shape

C1, also called the atlas. Does it have a body or spinous process?

C2, also called the axis. Has a distinct odontoid process (also called the dens)

☐ C2, C6 often have a bifid spinous process
☐ b. Thoracic vertebrae, T1 - T12

Typically have a long spinous process and a heart-shaped vertebral foramen. Articular facets present on transverse processes (articulate with ribs). Articular facets (or demifacets) present the body (articulate with ribs)

☐ **c. Lumbar vertebrae, L1 – L5**

Typically have a large stout/robust body, and a blunt spinous process d. Sacrum, made up of 5 fused sacral vertebrae S1, S5

FUSED VERTEBRAL BODES	AURICULAR SURFACE
transverse ridges	superior articular processes and facets
sacral promontory	sacral canal*
ala	anterior and posterior sacral foramina

e. Coccyx, made up of 3–5 fused coccygeal vertebrae

☐ **3. Identifying the bones of the thorax**

☐ **A. Ribs (costa) All paired. Numbered 1, 12 from superior to inferior**

GENERAL STRUCTURE OF THE TYPICAL RIB	THREE TYPES
head	1. vertebrosternal ribs 1–7
neck	All possess cartilage attachment to sternum
articular tubercle	
angle of rib	2. vertebrochondral ribs 8, 9, 10
shaft (body)	These attach to cartilage superior to it
costal groove	
costal cartilage*	3. vertebral or floating ribs
	These have no cartilage attachment

☐ **B. Sternum**

☐ manubrium jugular (suprasternal) notch
☐ body clavicular notch
☐ xiphoid process

Complete the following questions about the axial skeleton

What are the unique structural features of the atlas?

What are the differences between a vertebrosternal, vertebrochondral and vertebral rib?

List the five major regions of the vertebral column

List the four distinct curvatures that give the vertebral column an S-shape

WORKSHEET 2.4

The Axial Skeleton II: The Skull

- [ ] **Identifying the bones of the skull**

The Cranium (cranial bones, bone markings and features)

Frontal bone (usually fuses into a single bone during development)

- [ ] supraorbital foramen (or notch)
- [ ] glabella
- [ ] frontal sinuses*
- [ ] Parietal bone (paired)
- [ ] sagittal suture squamous suture
- [ ] coronal suture lambdoidal (lambdoid) suture
- [ ] Occipital bone (single)
- [ ] foramen magnum lambdoidal suture
- [ ] occipital condyles hypoglossal canal
- [ ] external occipital protuberance
- [ ] Temporal bones (paired)

squamous region:

- [ ] zygomatic process of the temporal bone
- [ ] mandibular fossa
- [ ] squamosal suture

tympanic region:

external auditory meatus (external acoustic meatus)

- [ ] **styloid process**

mastoid region:

- [ ] mastoid process
- [ ] stylomastoid foramen*

petrous region:

internal auditory meatus (internal acoustic meatus)

- [ ] jugular foramen
- [ ] carotid foramen (carotid canal)
- [ ] Sphenoid bone (single)
- [ ] body foramen rotundum
- [ ] sella turcica foramen ovale
- [ ] greater wings & lesser wings foramen spinosum
- [ ] optic canal (optic foramen) foramen lacerum
- [ ] superior orbital fissure sphenoid sinus

Ethmoid bone (single) cribriform plate olfactory foramina crista galli ethmoid sinuses perpendicular plate superior* and middle nasal conchae

- [ ] The Facial bones
- [ ] Nasal bones (paired)

Maxilla (paired) plural is maxillae or maxillary bones palatine process of the maxilla incisive fossa or foramen infraorbital foramen alveolar margin (alveolar ridge) maxillary sinus

Mandible (single) body of the mandible mandibular angle ramus (plural is rami) coronoid process mandibular condyle

- [ ] mandibular foramen
- [ ] mental foramen
- [ ] alveolar margin (alveolar ridge)

Lacrimal bones (paired) lacrimal canal or sulcus

Zygomatic bones (paired) temporal process of the zygomatic bone

- [ ] Vomer (single)
- [ ] Palatine bones (paired)

Inferior nasal conchae (paired) concha is singular

- [ ] Auditory ossicles (all paired)
- [ ] Malleus hammer
- [ ] Incus anvil
- [ ] Stapes stirrup
- [ ] Special Parts of the Skull
- [ ] Pterion
- [ ] Orbits
- [ ] Nasal Cavity
- [ ] Hyoid bone (not part of the skull)
- [ ] Superior, middle and inferior meatuses

Paranasal Sinuses, frontal sinus, sphenoid sinus, maxillary sinus

Complete the following questions about the axial skeleton

- [ ] **Name the bones that contribute to the orbit**

What is clinically important about the pterion?

What structures contribute to the zygomatic arch?

WORKSHEET 2.5

The Joints Part I, The Fibrous and Cartilaginous Joints

Classification of Joints by Structure and Function

Joints are classified by structure and function

Structural Classifications Functional Classifications (mobility)

- [ ] **Fibrous joints Synarthroses (immovable)**

Cartilaginous joints Amphiarthroses (slightly movable)

- [ ] **Synovial joints Diarthroses (freely movable)**

Use your textbook (Chapter 9) to complete the following table

STRUCTURAL CLASS	TYPES	STRUCTURAL CHARACTERISTICS OF JOINT	FUNCTIONAL CLASS (AMOUNT OF MOVEMENT PERMITTED AT JOINT)
Fibrous	1. Suture		
	2		
	3		
Cartilaginous	1. Synchondrosis		
	2. Symphysis		

- [ ] **2. Complete the following questions**
- [ ] **Define a joint**

Where in the body do find sutures? How mobile are these joints?

Name two syndesmoses. How mobile is each? Why?

Where in the body do you find gomphoses?

Where in the body do you find synchondroses? How mobile are these joints?

Where in the body do you find symphyses? How mobile are these joints?

The Joints Part II, The Synovial Joints

Classification of Joints by Structure and Function

Joints are classified by structure and function

Structural Classifications Functional Classifications (mobility)

- [ ] **Fibrous joints Synarthroses (immovable)**

Cartilaginous joints Amphiarthroses (slightly movable)

- [ ] **Synovial joints Diarthroses (freely movable)**

Use your textbook (Chapter 9, pages 205–206) to complete the following table

STRUCTURAL CLASS	TYPES	STRUCTURAL CHARACTERISTICS OF JOINT	FUNCTIONAL CLASS (AMOUNT OF MOVEMENT PERMITTED AT JOINT)
Synovial	1		
	2. Hinge		
	3		
	4		
	5		
	6. Ball-and-socket		

- [ ] **Movements allowed by synovial joints**

With the help of your lab partner demonstrate the following joint movements

- [ ] Gliding movements
- [ ] Rotation
- [ ] Angular movements

FLEXION	EXTENSION	ANKLE JOINT ONLY:	ANKLE JOINT ONLY:
abduction	adduction		
		dorsiflexion	plantarflexion
circumduction		eversion	inversion

- [ ] **Special movements**

PRONATION	SUPINATION
protraction	retraction
elevation	depression

WORKSHEET 2.6

The Structure of Bones and Muscles

- [ ] **1. Gross anatomy of a long bone**

- [ ] compact bone (tissue)
- [ ] epiphysis
- [ ] medullary cavity
- [ ] endosteum*
- [ ] diaphysis
- [ ] trabeculae (structure)
- [ ] articular cartilage*
- [ ] spongy bone (tissue)
- [ ] epiphyseal line
- [ ] periosteum*

- [ ] **2. Microscopic anatomy of a bone**

OSTEON	LAMELLA (PLURAL: LAMELLAE)
central (Haversian) canal	interstitial lamellae
perforating (Volkmann's) canal	circumferential lamellae
endosteum*	canaliculus (plural: canaliculi)
periosteum	lacuna (plural: lacunae)
perforating (Sharpey's) fibers	trabecula (plural: trabeculae)
collagen fibers (between lamellae)	red marrow

- [ ] **3. Microscopic anatomy of a skeletal muscle fiber / cell**

SARCOMERE	MYOFIBRIL
I-band	sarcolemma
A-band	sarcoplasm
Z-disc	sarcoplasmic reticulum
M-line H-zone	t-tubule endomysium (peripheral) nucleus

Use the models in lab and your textbook (Chapter 10) to draw a longitudinal view of a skeletal muscle fiber and label each of the structures listed above:

UNIT 3

Muscle, the Heart and the Vessels

MUSCLE, THE HEART, THE VESSELS AND THE NERVES OF THE UPPER LIMB

Muscle from the sarcomere out, then the heart, the vessels and the nerves of the chest and upper limb.

☐ **Drill these on the models**

LAB SPRINTS FOR THIS UNIT

Muscle Microanatomy
Head and Neck Muscles
Chest and Anterior Brachium Muscles
Posterior Thorax, Shoulder, and Posterior Brachium
Antebrachium Muscles
The Heart
Cardiac Conduction
Blood Vessels

WORKSHEET 3.1

Anatomy of the Chest and Anterior Brachium

☐ **Skeletal Muscle System**

Use Chapter 11 in your textbook to complete the following table. You are responsible for identifying, naming and knowing the origin(s), insertion(s) and major action(s) of the following muscles

MUSCLE	ORIGIN(S)	INSERTION(S)	ACTION(S)
Pectoralis major			
Pectoralis minor			
Serratus anterior			
Subscapularis*			
Coracobrachialis			
Biceps brachii Long head Short head			
Brachialis			

* Subscapularis is 1 of 4 muscles that contribute to the rotator cuff

☐ **2. Cardiovascular System. Use chapter 20 to identify the following on the specimens and models. All of these structures are paired. Include the word “right” or “left” when naming vessels and nerves**

ARTERIES
Subclavian artery, axillary artery, brachial artery
Deep veins
Subclavian vein, axillary vein, brachial vein
Superficial veins
Basilic vein, cephalic vein, median cubital vein

☐ **3. Nervous System. Use chapter 14 to identify the following peripheral nerves**

Musculocutaneous nerve, median nerve, ulnar nerve

The musculocutaneous nerve innervates the biceps brachii, brachialis, and coracobrachialis muscles. These muscles are contained within which muscle compartment? ____________________. The musculocutaneous nerve also innervates the skin of the lateral forearm

WORKSHEET 3.2

Anatomy of the Posterior Thorax, Shoulder and Posterior Brachium

☐ Skeletal Muscle System

Use pages 296 - 301 in your textbook to complete the following table. You are responsible for identifying, naming and knowing the origin(s), insertion(s) and major action(s) of the following muscles

MUSCLE	ORIGIN(S)	INSERTION(S)	ACTION(S)
Superficial muscles of the back and shoulder Trapezius			
Latissimus dorsi			
Teres major			
Deltoid			
Deeper muscles of the back and shoulder Rhomboid minor			
Rhomboid major			
Levator scapulae			
Supraspinatus*			
Infraspinatus*			
Teres minor*			
Muscles of the posterior brachium Triceps brachii			
Long head Lateral head Medial head			

* These muscles (and subscapularis) form the rotator cuff which is a critical stabilizing element for the glenohumeral joint (shoulder joint). Thus, one action for each of these muscles is to stabilize the shoulder joint. Examine the rotator cuff muscles in the cadaver and see if you can determine where the tendons incompletely surround the head of the humerus. What clinical significance does this have?

☐ Peripheral nerves

Radial nerve (view in the proximal posterior arm between long and lateral heads of triceps brachii) innervates the long, lateral, and medial heads of the triceps brachii in the posterior brachial muscle compartment

WORKSHEET 3.3

Anatomy of the Anterior Antebrachium (forearm)

☐ **Skeletal Muscle System**

Use Chapter 11 in your textbook to complete the following table. You are responsible for identifying, naming and knowing the origin(s), insertion(s) and major action(s) of the following muscles

MUSCLE	ORIGIN(S)	INSERTION(S)	ACTION(S)
Superficial layer Pronator teres			
Flexor carpi radialis			
Palmaris longus (may not be present)			
Flexor carpi ulnaris			
Intermediate layer Flexor digitorum superficialis			
Deep layer Flexor digitorum profundus			
Flexor pollicis longus			
Pronator quadratus			

☐ **2. Cardiovascular System. Use chapter 20 to identify the following on the specimens and models. All of these structures are paired. Include the word “right” or “left” when naming vessels and nerves**

ARTERIES
brachial artery, radial artery, ulnar artery
Superficial veins
Basilic vein, cephalic vein

☐ **3. Nervous System. Use chapter 14 to identify the following peripheral nerves**

☐ **Peripheral nerves**

Ulnar nerve, median nerve, musculocutaneous nerve, radial nerve

The ulnar nerve innervates flexor carpi ulnaris and flexor digitorum profundus (in part) and intrinsic hand muscles. It also innervates the skin of the medial hand and of digit 5 and parts of digits 3 and 4

The median nerve innervates most of the muscles of the anterior forearm muscle compartment (pronator teres, flexor carpi ulnaris, palmaris longus, flexor digitorum superficialis, flexor digitorum profundus (in part) flexor pollicis longus, pronator quadratus) as well as intrinsic hand muscles. It also innervates the skin of the palm and parts of digits 1–4

WORKSHEET 3.4

Anatomy of the Posterior Antebrachium (forearm)

☐ Skeletal Muscle System

Use chapter 11 in your textbook to complete the following table. You are responsible for identifying, naming and knowing the origin(s), insertion(s) and major action(s) of the following muscles

MUSCLE	ORIGIN(S)	INSERTION(S)	ACTION(S)
Brachioradialis			
Extensor carpi radialis longus			
Extensor carpi radialis brevis			
Extensor digitorum			
Extensor digiti minimi			
Extensor carpi ulnaris			
Abductor pollicis longus			
Extensor policis brevis			
Extensor pollicis longus			
Extensor indicis			

☐ Peripheral nerves

Radial nerve (easily viewed in the proximal posterior arm between long and lateral heads of triceps brachii)

The radial nerve innervates the: brachioradialis, extensor carpi radialis longus and brevis, extensor carpi ulnaris, extensor digitorum, extensor digiti minimi, abductor pollicis longus, extensor pollicis longus and brevis, and extensor indicis

What is the name of the muscle compartment that houses most of these muscles? ____________________

The radial nerve also innervates the skin of the dorsum of the hand and parts of digits 1–4

WORKSHEET 3.5

Anatomy of the Thorax – The Anatomy of Heart

☐ **Gross Anatomy of the Human Heart**

Using Chapter 19 in your textbook examine and identify the gross anatomy of the heart and associated structures on the specimens and models provided. Identify the following:

THORACIC CAVITY, MEDIASTINUM, PERICARDIAL CAVITY	MAJOR SYSTEMIC VEINS
Fibrous pericardium	Inferior vena cava (empties into RA)
Serous pericardium (parietal & visceral layers) Epicardium, myocardium, endocardium	Superior vena cava (empties into RA) R & L brachiocephalic veins R & L subclavian veins R & L internal jugular veins
R & L Atria (singular: atrium)	Coronary sinus (empties into RA)
R & L auricles Interatrial septum Pectinate muscles	Vessels that exit the R ventricle Pulmonary trunk R & L pulmonary arteries
R & L Ventricles	
Interventricular septum	Vessels that empty into the L atrium
R atrioventricular valve (tricuspid valve)	R & L pulmonary veins (4 total)
Pulmonary semilunar valve	
L atrioventricular valve (bicuspid / mitral valve)	Major Systemic Arteries*
Aortic semilunar valve	Ascending aorta becomes the aortic arch
Chordae tendineae	Three major branches of the aortic arch:
Papillary muscles	Brachiocephalic trunk
Trabeculae carneae	L common carotid artery
	L subclavian artery
Blood supply to the myocardium	* Although no longer an artery
R & L coronary arteries	identify the ligamentum arteriosum
Anterior interventricular artery also called	
the L anterior descending artery (LAD)	
Circumflex branch of the L coronary artery	

☐ **1.1 Compare the diameter and wall thickness of the superior vena cava and the aorta. Which is larger?**

Which has thicker walls? Why do you suppose these differences exist?

☐ **1.2 Compare the wall thickness of the right and left ventricles. Which is thicker? Why?**

The layers of the pericardium and of the heart wall

Draw a diagram of the layers of the pericardium. Draw a diagram of the layers of the heart wall

☐ **Complete the following short answer questions**

☐ **3.1 Describe the structure and location of the atrioventricular valves. Describe how these**

valves work to prevent the backflow of blood in the heart

☐ **3.2 Describe the structure and location of the semilunar valves. Describe how these**

valves work to prevent the backflow of blood in the heart

☐ **3.3 Trace one drop of blood through all heart chambers and heart valves, and through the basic vascular circuits from the time it enters the right atrium until it enters the right atrium again**

☐ **3.4 What is the Cardiac Conduction System? List the components of the Cardiac Conduction System**

Define the following terms:

- [ ] systole -
- [ ] diastole -
- [ ] angina pectoris -
- [ ] myocardial infarction

UNIT 4

Limb Muscles, Respiratory and Digestive

THE LIMB MUSCLES, THE RESPIRATORY TRACT AND THE DIGESTIVE TRACT

The muscles of the thigh and leg, then the respiratory tract and the digestive tract.

- [ ] **Drill these on the models**

LAB SPRINTS FOR THIS UNIT

Thigh Muscles
Leg Muscles
Respiratory System
Alimentary Canal
Accessory Digestive Organs
Abdominal Wall and Peritoneal Cavity

WORKSHEET 4.1A

Anterior, Medial, and Lateral Thigh

- [ ] **Skeletal Muscle System**

Study the following muscles on the specimens and models. Be able to name and identify each muscle, and describe the origin(s), insertion(s) and action(s)

MUSCLE	ACTION(S)	ORIGIN(S)	INSERTION
Iliopsoas			
Iliacus			
Psoas major			
Quadriceps femoris			
Rectus femoris			
Vastus lateralis			
Vastus intermedius			
Vastus medialis			
Sartorius			
Gracilis			
Adductor magnus			
Adductor longus			
Adductor brevis			
Pectineus			
Tensor fasciae latae			

- [ ] **2. Nervous System – Identify the following branch of the lumbar plexus**
- [ ] **Femoral nerve**
- [ ] **3. Cardiovascular System (Use right/left and artery/vein). These vessels are all paired**
- [ ] **Femoral artery**

Femoral vein (a deep vein), Great saphenous vein (a superficial vein)

WORKSHEET 4.1B

Gluteal Region and Posterior Thigh

Skeletal Muscle System - Muscles of the posterior hip and thigh

Study the following muscles on the specimens and models. Be able to name and identify each muscle, and describe the origin(s), insertion(s) and action(s)

MUSCLE	ACTION(S)	ORIGIN(S)	INSERTION
Gluteus maximus			
Gluteus medius			
Hamstrings			
Biceps femoris			
Semitendinosus			
Semimembranosus			

☐ **2. Nervous System – Identify the following branches of the sacral plexus:**

Sciatic nerve (which later branches into the tibial nerve and common fibular nerve)

☐ **3. Cardiovascular System (Use right/left and artery/vein). These vessels are all paired**

☐ Femoral artery ☐ Femoral vein (a deep vein) ☐ Great saphenous vein (a superficial vein)

WORKSHEET 4.2

The Leg

☐ **Skeletal Muscle System – Muscles of the leg**

Study the following muscles on the specimens and models. Be able to name and identify each muscle, and describe the origin(s), insertion(s) and action(s)

MUSCLE	ACTION(S)	ORIGIN(S)	INSERTION
Tibialis anterior			
Extensor hallucis longus			
Extensor digitorum longus			
*Fibularis tertius			
*Fibularis brevis			
*Fibularis longus			
Tibialis posterior			
Flexor digitorum longus			
Flexor hallucis longus			
Gastrocnemius			
Soleus			
Plantaris			

*You may see Peroneus used instead. Fibularis and Peroneus are synonyms

☐ **2. Nervous System – Identify the following:**

☐ Tibial nerve
☐ Common fibular nerve

☐ **3. Cardiovascular System (Use right/left and artery/vein). These vessels are all paired**

☐ Posterior tibial artery
☐ Great saphenous vein (a superficial vein)

WORKSHEET 4.3

Thoracic Anatomy, Respiratory System, selected vessels and associated structures

We typically delay this material to match lecture material

Anatomy of the Thorax - Respiratory System Anatomy

Using figures in your textbook examine and identify the gross anatomy of the respiratory system on the specimens and models provided. You are responsible for these structures:

THORACIC CAVITY	HILUM OF THE R AND OF THE L LUNG
R and L pleural cavities	L & R pulmonary arteries
Diaphragm	L & R pulmonary veins
External intercostals, internal intercostals	
Mediastinum	L & R primary (main) bronchi
Trachea	Secondary (lobar) bronchi
Tracheal cartilages Carina	Tertiary (segmental) bronchi
Lobes of the R. lung (sup., mid., inf.)	
Lobes of the L. lung (sup., inf.)	Parietal pleura and visceral pleura

Anatomy of the Thorax, Additional vessels and structures

- ☐ Brachiocephalic trunk
- ☐ R and L brachiocephalic veins
- ☐ R and L subclavian arteries
- ☐ R and L subclavian veins
- ☐ R and L common carotid arteries
- ☐ R and L internal jugular veins
- ☐ R and L phrenic nerves
- ☐ Esophagus

☐ **Anatomy of the Thorax, Study questions**

Which respiratory structures are part of the conducting zone of the respiratory system?

Why is it important that the trachea be reinforced with c-shaped cartilaginous rings? Why is it advantageous that these rings are incomplete posteriorly?

Distinguish between alveolus, alveoli, alveolar duct and alveolar sac

☐ **What structures make up the respiratory zone**

Describe the structure and function of the alveoli and the respiratory membrane?

Is the “bronchi” singular or plural? What about “bronchus”?

WORKSHEET 4.4

Anterior Cervical (Neck) Region

Skeletal Muscle System - Muscles of the anterior cervical region (neck)

Study the following muscles on the specimens and models. Be able to name and identify each muscle, and describe the origin(s), insertion(s) and action(s)

MUSCLE	ACTION	ORIGIN	INSERTION
Platysma			
Sternocleidomastoid			
Omohyoid			
Sternohyoid			
Sternothyroid			
Thyrohyoid			

- ☐ **2. Nervous System**
- ☐ **Identify the vagus nerve (cranial nerve X)**
- ☐ **3. Vascular System (Use right/left and artery/vein when naming vessels.) These vessels are all paired**
- ☐ **Identify the: common carotid artery**

Identify the: vertebral artery (not exposed on cadavers - use a model or diagram)

- ☐ **Identify the: internal jugular vein**
- ☐ **5. Respiratory System Identify the following structures on models, prosections, or cadavers**

☐ Larynx ☐ Cricoid cartilage ☐ Vocal cords (vocal folds) ☐ Trachea
☐ Thyroid cartilage ☐ Cricothyroid ligament ☐ Glottis
☐ Larygeal prominence ☐ Arytenoid cartilages ☐ Epiglottis

- ☐ **6. Digestive System**
- ☐ **Esophagus, Submandibular glands**
- ☐ **7. Endocrine System**
- ☐ **Thyroid gland**

WORKSHEET 4.5

Abdominal Region I

☐ **Muscular System – Muscles of the abdomen**

Study the following muscles on the specimens and models. Be able to name and identify each muscle, and describe the action(s)

MUSCLE	ACTION(S)	ORIGIN(S)	INSERTION(S)
Rectus abdominis			
External abdominal oblique			
Internal abdominal oblique			
Transversus abdominis			

☐ **2. Ventral body cavities and serous membranes, and omenta. Identify the following structures on cadavers, prosections, diagrams, and models**

Abdominal cavity, pelvic cavity, peritoneal cavity

☐ Parietal and visceral peritoneum ☐ Greater omentum ☐ Lesser omentum

☐ **3. Digestive System, Gastrointestinal tract. Identify the following structures on cadavers, prosections, diagrams, and models**

ESOPHAGUS	LARGE INTESTINE (LARGE BOWEL)
	Cecum
Stomach	Vermiform appendix
Greater and lesser curvatures	Ascending colon
Gastric rugae	Hepatic flexure (also called the right colic flexure)
Cardiac region	Transverse colon
Fundus	Splenic flexure (also called the left colic flexure)
Body	Descending colon
Pyloric region	Sigmoid colon
Pyloric sphincter	Rectum
	Teniae coli
Small intestine (small bowel)	Haustra (singular: haustrum)
The mesentery	Anal canal
Duodenum	
Major duodenal papilla	
Jejunum	
Ileum	
Ileocecal valve	

WORKSHEET 4.6

Abdominal Region II

☐ **3. Digestive System, Accessory organs and associated structures. Identify the following structures on cadavers, prosections, diagrams, and models**

PANCREAS (THE PANCREAS IS ALSO AN ENDOCRINE ORGAN)	GALL BLADDER
Main pancreatic duct	
Accessory pancreatic duct	Biliary tree
Hepatopancreatic ampulla	R and L hepatic ducts
Major duodenal papilla	Common hepatic duct
	Cystic duct
Liver	Common bile duct
Right lobe	
Left lobe	
Quadrate lobe	
Caudate lobe	
Falciform ligament	
Ligamentum teres (also called the round ligament)	

☐ **4. Lymphatic System. Identify the following on cadavers, prosections, diagrams, and models**

☐ **Spleen**

☐ **5. Cardiovascular System. Identify the following structures on cadavers, prosections, diagrams, and models**

IDENTIFY THE FOLLOWING ARTERIES:	IDENTIFY THE FOLLOWING VEINS:
Abdominal aorta	Inferior vena cava
	R and L common iliac veins
Celiac trunk and its branches	
Left gastric artery	The hepatic portal vein and its major contributing veins
Splenic artery	Hepatic portal vein
Common hepatic artery	Splenic vein
	Inferior mesenteric vein
Superior mesenteric artery	Superior mesenteric vein
Inferior mesenteric artery	
R and L common iliac arteries	

UNIT 5

Urinary, Reproductive and the Nervous System

THE URINARY AND REPRODUCTIVE TRACTS, AND THE NERVOUS SYSTEM

The urinary and reproductive tracts, then the brain, the cord, the cranial nerves and the plexuses.

- [ ] **Drill these on the models**

LAB SPRINTS FOR THIS UNIT

Urinary System
Male Reproductive System
Female Reproductive System
Meninges, Ventricles, and CSF
Spinal Cord
The Brainstem
Cranial Nerves
Spinal Nerves and Plexuses

WORKSHEET 5.1

Pelvic Anatomy I – Urinary System and Selected Skeletal Muscles

☐ Gross Anatomy of the Urinary System Organs

Identify the following on models and specimens:

- [ ] Renal capsule
- [ ] Major Calyx
- [ ] Renal cortex
- [ ] Minor Calyx
- [ ] Renal medulla
- [ ] Renal Pelvis
- [ ] Renal column
- [ ] Ureter
- [ ] Renal pyramid
- [ ] Urinary Bladder
- [ ] Urethra

☐ Vasculature of the Kidney

Examine the path of blood vessels through the kidneys on the models and your text

Identify the following:

- [ ] Renal artery
- [ ] Renal vein
- [ ] Arcuate artery
- [ ] Arcuate vein
- [ ] Cortical radiate artery
- [ ] Cortical radiate vein
- [ ] Afferent arteriole
- [ ] Efferent arteriole

☐ Renal Corpuscle

Examine and identify the renal corpuscle: glomerulus, and glomerular capsule (also called Bowman's capsule) on the models and in your text. Identify the following:

PARIETAL LAYER OF GLOMERULAR CAPSULE	VISCERAL LAYER OF GLOMERULAR CAPSULE

The Renal Tubule. Examine and identify the anatomy of the renal tubule (also called the uriniferous tubule) on the models and in your text. Identify the following:

PROXIMAL CONVOLUTED TUBULE (PCT)	COLLECTING DUCT
Nephron loop (also called the loop of Henle)	Papillary duct (distal portion of collecting duct)
Distal convoluted tubule (DCT)	
	Juxtaglomerular apparatus
	Macula densa cells
	Granular cells

☐ 5. Complete the Following Questions:

What are the three major mechanisms by which the kidneys form urine?

What structures make up the juxtaglomerular apparatus?

What is the function of each structure that contributes to the juxtaglomerular apparatus?

What structures make up a renal corpuscle?

What is micturition?

The kidneys are retroperitoneal organs. What does this mean?

WORKSHEET 5.2

Pelvic Anatomy II, Male Reproductive System and Selected Pelvic Structures

☐ Reproductive System, The Male

Study the following on the specimens and models. Be able to name and identify each

SCROTUM	SEMINAL VESICLES
Testes (singular is testis)	Prostate gland
Epididymis	Bulbourethral glands (use a model or diagram)
Spermatic cord	Penis
Vas deferens (also called ductus deferens)	Corpus spongiosum
Testicular artery and vein	Corpora cavernosa (singular is corpus cavernosum)
Cremaster muscle	Glans penis
Pampiniform plexus	Prepuce
Inguinal canal	Urethra
Ejaculatory duct	Note: the male urethra has 3 regions:
	prostatic urethra
	membranous urethra
	spongy urethra (also called the penile urethra)

Circulatory System, Selected blood vessels of the pelvic region

Study the following on the specimens and models. Be able to name and identify each

ARTERIES
Internal iliac arteries
External iliac arteries
Medial umbilical ligaments (what structure was this in utero?)
Veins
Internal iliac veins
External iliac veins

☐ Complete the following short answer questions

Why are the testes located in the scrotum rather than inside the ventral body cavity?

What structures compose the spermatic cord?

How might enlargement of the prostate gland interfere with urination or reproductive function in men?

Trace the path of the male reproductive tract from the seminiferous tubules of the testis to the outside of the body. Include in your discussion all of the related glands

WORKSHEET 5.3

Pelvic Anatomy III, Female Reproductive System

☐ **Reproductive System, The Female**

Study the following on the specimens and models. Be able to name and identify each. Refer to the figures in your text

OVARY	UTERUS
	Fundus
Ovarian ligament	Body
Ovarian artery & vein	Cervix
	Endometrium
Serous Membranes	Myometrium
Broad ligament (of the uterus)	Perimetrium
Suspensory ligament (of the ovary)	
	Vagina
Fallopian tubes (also called uterine tubes or oviducts)	Vaginal orifice
Fimbriae	Fornix
Infundibulum	
Ampulla	Vulva (external genitalia)
Isthmus	Mons pubis
	Labia majora
Round ligament (of the uterus)	Labia minora
	Clitoris (with corpora cavernosa)
	Urethral orifice

Circulatory System, Selected blood vessels of the pelvic region

Study the following on the specimens and models. Be able to name and identify each. Refer to the figures in your text

ARTERIES
Internal iliac arteries
External iliac arteries
Medial umbilical ligaments
Veins
Internal iliac veins
External iliac veins

☐ **Complete the following short answer questions**

What space is the egg released into upon ovulation? How does it arrive in the uterus?

What would happen if an egg implanted in a fallopian tube?

What is the difference between the vulva and the vagina?

WORKSHEET 5.4

Anatomy of the Nervous System I, The Spinal Cord and Spinal Nerves

☐ CNS -- Gross Anatomy of the Spinal Cord

Observe the gross specimens of the spinal cord found in the laboratory. Use text figures to identify the following:

- ☐ The meninges
- ☐ Dura mater
- ☐ Arachnoid mater
- ☐ Pia mater
- ☐ Cervical enlargement
- ☐ Lumbar enlargement
- ☐ Conus medullaris
- ☐ Cauda equina Filum terminale

CNS -- Cross-sectional View of the Spinal Cord

Using your textbook, models, and wet specimens in the lab to identify the following underlined structures visible from a cross-sectional view of the spinal cord:

- ☐ Ventral horn
- ☐ Lateral horn
- ☐ Dorsal horn
- ☐ Central canal
- ☐ Gray matter
- ☐ White matter
- ☐ Grey commissure

☐ PNS -- Anatomy of Spinal Nerves

Using your textbook, models, and wet specimens in the lab to identify the following underlined structures visible from a cross-sectional view of the spinal cord:

- ☐ Ventral root
- ☐ Dorsal ramus
- ☐ Grey ramus communicans
- ☐ Sympathetic chain ganglion
- ☐ Ventral ramus
- ☐ Dorsal root ganglion
- ☐ White ramus communicans
- ☐ Dorsal root
- ☐ Spinal nerve

☐ Spinal Cord and Spinal Nerve Review

Using a microscope slide showing a cross-sectional view of the spinal cord, a model showing the spinal cord and spinal nerves, and your textbook, draw and label a cross-sectional view of the spinal cord and spinal nerves including as many of the terms in sections 2 and 3 above as possible

☐ PNS -- Gross Anatomy of the Spinal Cord

Using your textbook and models, identify the following nerve plexuses:

CERVICAL PLEXUS	BRACHIAL PLEXUS	LUMBAR PLEXUS	SACRAL PLEXUS

WORKSHEET 5.5

Anatomy of the Nervous System II, Brain and Cranial Nerves

CNS -- Gross Anatomy of the Brain. Observe the brain using both the models and gross specimens found in the laboratory. Using your textbook. Identify the following:

CEREBRUM	MIDBRAIN
Frontal lobes	Corpora quadrigemina
Parietal lobes	Superior colliculi
Occipital lobes	Inferior colliculi
Temporal lobes	
Insula	Pons
Corpus callosum	Trigeminal nerve
Fornix	
Sulci (singular: sulcus)	Cerebellum
Gyri (singular: gyrus)	Arbor vitae (pattern of white matter)
Central sulcus	
Precentral gyrus (primary motor cortex)	Medulla oblongata
Postcentral gyrus (primary somatosensory cortex)	Medullary pyramids, most corticospinal
Lateral fissure	nerve fibers decussate here
Longitudinal fissure	
Transverse fissure	The Meninges
Olfactory bulbs and olfactory tracts	Dura mater
	Arachnoid mater
Diencephalon	Pia mater
Thalamus	
Hypothalamus	CSF production and circulation
Pineal body (pineal gland)	Choroid plexuses (in each ventricle)
Optic nerves	Lateral ventricles
Optic chiasma	Third ventricle
Mammillary bodies	Cerebral aqueduct
	Fourth ventricle
Pituitary gland	Superior sagittal sinus
	Sigmoid sinus

☐ **PNS -- The Cranial Nerves**

Observe the 12 paired cranial nerves of the brain using both the models and gross specimens found in the laboratory. Use your textbook to identify the following cranial nerves by name and Roman numeral. What organs do these nerves innervate? What types of signals do they carry?

NUMBER	NAME	TYPE	INNERVATED ORGANS
I	Olfactory nerve	S	
II	Optic nerve	S	
III	Oculomotor nerve	M	
IV	Trochlear nerve	M	
V	Trigeminal nerve	B	
VI	Abducens nerve	M	
VII	Facial nerve	B	
VIII	Vestibulocochlear nerve	S	
IX	Glossopharyngeal nerve	B	
X	Vagus nerve	B	
XI	Spinal accessory nerve	M	
XII	Hypoglossal nerve	M	

Anatomy Supplement

Anatomy of the Nervous System I, The Spinal Cord and Spinal Nerves

- [ ] **Cross-sectional view of the Spinal Cord**

Note that the spinal cord is part of the central nervous system or CNS

- [ ] **White Matter**

Consists of tracts of myelinated axons travelling to or from similar locations within the CNS

Examples: spinothalamic tract; corticospinal tract

- [ ] **Gray Matter**

Consists of cell bodies, dendrites, synapses, and some un-myelinated axons in the CNS

- [ ] **Ventral horns**

Cell bodies of somatic motor neurons are located here. The ventral horns are also called the anterior horns

- [ ] **Lateral horns**

Cell bodies of visceral (specifically sympathetic) motor neurons are located here

- [ ] **Dorsal horns**

Cell bodies of interneurons that process sensory information are located here. The dorsal horns are also called posterior gray horns

- [ ] **Gray commissure**

Gray matter consisting of axons that decussate (cross the midsagittal plane) so that the left half of the spinal cord can communicate with the right half of the spinal cord (and vice-versa)

- [ ] **Central canal**

Narrow (and often incomplete) tube in the center of the gray matter of the spinal cord that contains cerebrospinal fluid

- [ ] **Anatomy of Spinal Nerves**

Note that spinal nerves are part of the peripheral nervous System or PNS

31 pairs of spinal nerves emerge from each of the 31 segments of the spinal cord. Each spinal nerve communicates with the spinal cord via groups of tiny rootlets that converge distally to form two roots. More distally, these two roots converge to form the true spinal nerves

Dorsal root:

Consists entirely of axons that conduct afferent sensory information from the periphery toward the spinal cord

- [ ] **Dorsal root ganglion**

Contains the cell bodies of afferent sensory neurons (both somatic and visceral sensory neurons)

- [ ] **Located along each dorsal root**

Dendrites of these neurons are located in and around peripheral organs and axon terminals are inside the central nervous system

Ventral root: consists entirely of axons that conduct efferent motor information from the spinal cord toward the periphery

☐ **Spinal nerves**

Near the intervertebral foramina of the vertebrae, the dorsal and ventral roots converge to form true spinal nerves. Normally, spinal nerves emerge from the intervertebral foramina inferior to the vertebrae that have the same name (e.g. spinal nerve T11 emerges from the intervertebral foramen inferior to vertebra T11)

Exception: cervical spinal nerves emerge from the intervertebral foramina superior to the vertebrae that have the same name. (e.g. spinal nerve C3 emerges from the intervertebral foramen superior to vertebra C3). This occurs because there are eight cervical spinal nerves but only seven cervical vertebrae. Therefore, spinal nerve C8 emerges from the intervertebral foramen between vertebra C7 and T1

☐ Dorsal and ventral rami ☐ Dorsal ramus of spinal nerve

Just distal to where the axons of the dorsal and ventral roots fasciculate (converge) to form distinct spinal nerves, axons carrying sensory and motor information from or to parts of the body dorsal to the spinal cord branch posteriorly to form the dorsal ramus of the spinal nerve

Ventral ramus of spinal nerve:

The remainder of the spinal nerve innervates parts of the body anterior to the spinal cord and is called the ventral ramus of the spinal nerve

☐ **Autonomic fibers of spinal nerves**

Just distal to the location where spinal nerves T1 to L2 branch to form the dorsal and ventral rami, two additional rami (branches) emerge from the ventral ramus of the spinal nerve to communicate with the sympathetic chain ganglia. These are the rami communicantes (note that this term is plural)

Grey ramus communicans:

The more proximal of the two rami communicantes; contains the unmyelinated axons of post-ganglionic sympathetic neurons

White ramus communicans:

The more distal of the two rami communicantes; contains the myelinated axons of pre-ganglionic sympathetic neurons

Sympathetic chain ganglion (also called the sympathetic trunk ganglia or paravertebral ganglia)

Contains the cell bodies of some post-ganglionic sympathetic neurons

Approximately 20–23 pairs of sympathetic chain ganglia are located along the bodies of the cervical, thoracic, lumbar, and sacral vertebrae

Sympathetic chain ganglia communicate with each other via short nerves that extend from one ganglion to the next. These nerves together with the sympathetic chain ganglia are called the “sympathetic chain”

☐ **Nerve Plexuses**

Axons of the ventral rami of different spinal nerves branch and recombine with axons from the ventral rami of neighboring spinal nerves. These branching and interconnecting networks of nerves are called nerve plexuses

Four spinal nerve plexuses give rise to the major named peripheral nerves

☐ Cervical Plexus ☐ Gives rise to the phrenic nerves ☐ Brachial Plexus

Gives rise to the musculocutaneous, median, ulnar, radial, and axillary nerves

☐ **Lumbar Plexus**

Gives rise to the femoral and obturator nerves

☐ **Sacral Plexus**

Gives rise the sciatic nerve (i.e. the combined tibial and common fibular nerves)

There are also several autonomic nerve plexuses that supply the ventral body cavity. Examples include the cardiac plexus and the celiac plexus (also called the solar plexus)